



**Technological Institute of the Philippines
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FairShare: Finance Tracking Mobile Application for Personal and Group Expenses

AN ACTIVITY SUBMITTED IN PARTIAL FULFILLMENT
OF THE REQUIREMENT OF MOBILE COMPUTING

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Table of Contents

Problem Statement.....	3
Objectives.....	3
Specific Objectives.....	3
Target Users.....	4
1. The Social Spender (College Students / Barkada).....	4
2. The Reimbursable Pro (Working Professionals / Freelancers).....	4
3. The Digital Nomad (Expats / International Travelers).....	5
Features.....	5
Scope.....	6
Limitations.....	6

Problem Statement

People who are budget-conscious track their financial activity through financial apps. These apps let you log your monthly income and expenses. However, when people go on group events between friends (like road trips, vacations, out-of-country trips, etc.), some expenses that happen during these events are forgotten to be logged and leave the people with a feeling of money lost when, in reality, they have their money in the form of IOUs. these "context gaps," where users' ledgers are not updated on social spending, leaving users wondering where their money went.

Another problem that we see is the "awkwardness factor," where after the event, people forget that they owe someone some money, and the one who lends the money feels like they cannot get their money back from those that they lent some money to.

In addition to forgotten social expenses and awkward debt collection, an issue in group events is the complexity of unequal expense sharing. During trips or gatherings, not all participants consume the same item. One member may pay for gas, another for food, and another purchase personal items (e.g., clothes) that shouldn't be split equally.

Traditional budgeting applications use a simple "divide by N" approach, which results in inaccurate and unfair calculation cost distribution. This leads to financial confusion and manual computation burdens, especially when the receipts contain multiple sub-expense that involves different participants

And the last problem that we see is the problem of expense management for a group of people. Where the problem lies is in the manual calculation and tracking of expenses, especially if the expense arrangement is one person pays for one thing and another for another thing, and when someone ends up having to pay out of pocket for a sudden expense.

Objectives

To develop **FairShare**, a multi-pillar financial management application that bridges the gap between personal budgeting and social expense tracking, ensuring financial transparency and reducing social friction within groups.

Specific Objectives

1. **To Integrate Personal and Social Ledgers:** Create a unified system where users can log personal daily expenses and seamlessly "pivot" shared costs into group sessions, eliminating the "context gap" in their financial records.
2. **To Implement a Real-Time Collaborative "Lobby":** Utilize Firebase Realtime Database to provide a synchronized environment where group members can instantly view, add, and verify shared expenses during events.

3. **To Automate Complex Debt Simplification:** Develop a minimum debt path algorithm that calculates the most efficient way to settle group balances, reducing a "web of debt" into the fewest possible transactions.
4. **To Mitigate Social Friction via Automated Notifications:** Design a "Nudge" system using Firebase Cloud Messaging (FCM) that uses pre-written, polite, or humorous templates to remind peers of outstanding IOUs without the need for manual, awkward confrontation.
5. **To Provide Persistent Peer-to-Peer Tracking:** Establish a dedicated 1-on-1 Mode that maintains a rolling balance for long-term "friendly debts," ensuring that small, recurring loans between individuals are never forgotten.

Target Users

1. The Social Spender (College Students / Barkada)

This user is the heart of the "Social Friction" problem. They deal with high-frequency, low-value transactions where "asking for money" feels petty but necessary.

- **The Vibe:** Informal, mobile-first, and budget-conscious.
- **FairShare Interaction:**
 - **Group Mode:** Used for weekend "Inuman," road trips, or group study snacks.
 - **IOU Mode:** Tracking ₱50 coffee or ₱100 fare loans.
- **Refined Needs:**
 - **Gamified Nudges:** Instead of "Pay me back," the nudge could be, "Round 2 is on you if you don't settle this ₱50!"
 - **Social Visibility:** A "Hall of Fame" (or Shame) showing who is the fastest to settle their debts.

2. The Reimbursable Pro (Working Professionals / Freelancers)

This user needs to prove where their money went. They aren't just splitting a bill; they are managing a "work vs. life" balance.

- **The Vibe:** Organized, efficiency-driven, and hates manual data entry.
- **FairShare Interaction:**
 - **Personal Tracker:** Essential for separating personal spending from project-related expenses.
 - **Group Mode:** Used for "Team Lunch" or "Office Pitch" contributions.
- **Refined Needs:**
 - **OCR Receipt Scanner:** An audit-focused tool tailored specifically for corporate reimbursement. The ability to snap a photo of a BIR official receipt and have the app auto-extract the total and tax for the Personal Tracker, streamlining the logging of company dinners and official work expenses.

- **One-Tap Export:** A "Generate Report" button that creates a PDF/CSV for the HR department or their personal accountant.
- **Corporate Silos:** Multiple ledgers so their "Client A" expenses don't mix with their "Friday Night Drinks" expenses.

3. The Digital Nomad (Expats / International Travelers)

This user deals with "mathematical friction" on a global scale. They are often confused by exchange rates and conversion fees.

- **The Vibe:** Global, tech-savvy, and precision-oriented.
- **FairShare Interaction:**
 - **IOU Mode:** Tracking debts across borders (e.g., "I paid in USD, you pay me back in PHP").
- **Refined Needs:**
 - **Live FX API:** Integrating a real-time exchange rate (e.g., via ExchangeRate-API) so the 1-on-1 mode updates the "Net Owe" as currency values shift.
 - **Smart Rounding:** In countries like the Philippines, where small centavos are rarely used in cash, the app should offer a "Round to nearest ₱1 or ₱5" toggle.
 - **Cross-Border Remittance Links:** Providing info/links for Wise or Binance P2P for international settlement.

Features

1. User Account Management

- Makes account creation possible and includes the social aspect of the application.

2. Personal Finance Tracker

- Focuses on logging and managing daily expenses for the self and allows for preset and customizable tags to create custom labels on items spent. For professional users requiring official auditing, this module includes an OCR receipt scanner specifically tailored for extracting data from BIR-official receipts for corporate reimbursement and company dinner tracking.

3. Group Mode Finance Tracker

- Designed for payments shared among groups to quickly visualize the amount to pay each user for the occasion, minimizing the issues that come with following up on loan payments. This feature will support both "equal split" and "itemized split" methods, allowing a single large expense to be distributed fairly based on exactly who participated in each sub-expense.

4. IOU Tracker and Pinger

- Manages or creates alerts for payment tracking between users.

Scope

The application will be developed with the following aspects considered:

- **Core Functionalities:** Application features will allow for tracking expenses, be they personal or by groups. This will be made possible by implementing OCR for receipt scanning, IOU Tracker and Pinger for notifications, and a UI view to facilitate all user interactions within the system. Furthermore, specific features such as the Personal Finance Tracker will be accessible even without network connectivity but will need the internet to sync it to the database.
- **Target Users:** Designed for a diverse user population, ranging from teens to late adults, with a focus on individuals managing personal budgets, participating in peer-to-peer lending, or managing group funds.
- **Platform:** The application will be developed natively for mobile cellphones using the Android OS.

Limitations

The application will have constraints on:

- **Features:** Payment gateways will not be integrated into the system, as it will not be handling real-time financial transactions with other banking services or applications. In addition to this, specific features such as the Group Finance Tracker will require network connectivity to sync data across all users within a group.
- **Excluded Platforms:** FairShare will not be developed as a web application or for PC devices.
- **User Limitations:** In relation to the platform's aspect, users will not be able to access the mobile system through web browsers or PC devices—mobile devices with other operating systems will also be unable to use it.
- **Language Support:** The mobile application will only be available in English.